

HANDS-ON 8 [Advanced]

HTTP Client – API Integration, Observables & Interceptors

Aim:

To integrate Angular applications with REST APIs using HttpClient, perform CRUD operations, use RxJS operators for data handling, and implement HTTP Interceptors for authentication, loading, and error handling.

Task 1: Replace Service Data with HttpClient Calls

Aim:

To replace hardcoded course data with live API data using Angular HttpClient.

Steps:

- Install and configure JSON Server.
- Import HttpClientModule (or provideHttpClient()).
- Inject HttpClient into CourseService.
- Implement GET methods (getCourses(), getCourseById()).
- Subscribe to API data in CourseListComponent.
- Implement POST (createCourse()).
- Implement PUT (updateCourse()).
- Implement DELETE (deleteCourse()).
- Test all CRUD operations.

Task 2: RxJS Operators & Error Handling

Aim:

To transform API responses and handle HTTP errors using RxJS operators.

Steps:

- Use map() to transform course data.
- Use catchError() for error handling.
- Use tap() for logging.
- Use retry(2) for retrying failed requests.
- Use switchMap() for dependent API calls.

Task 3: HTTP Interceptors

Aim:

To implement HTTP Interceptors for authentication, loading, and global error handling.

Steps:

- Create an Authentication Interceptor.
- Add Authorization headers to requests.
- Register the interceptor in app.config.ts.
- Create an Error Interceptor.
- Handle 401 and 500 HTTP errors.
- Create a Loading Interceptor.
- Display a loading spinner using LoadingService.

Outputs:

```
student-course-portal > src > app > pages > course-list > TS course-list.ts > ...
1  import { Component, OnInit } from '@angular/core';
2  import { CommonModule } from '@angular/common';
3  import { FormsModule } from '@angular/forms';
4  import { ActivatedRoute, Router } from '@angular/router';
5
6  import { CourseCard } from '../../components/course-card/course-card';
7  import { CourseService } from '../../services/course';
8  import { Course } from '../../models/course.model';
9
10 @Component({
11   selector: 'app-course-list',
12   standalone: true,
13   imports: [
14     CommonModule,
15     FormsModule,
16     CourseCard
17   ],
18   templateUrl: './course-list.html',
19   styleUrls: ['./course-list.css']
20 })
21 export class CourseList implements OnInit {
22
23   courses: Course[] = [];
24   searchTerm = '';
25 }
```

styles.css | styles | 95 bytes | Initial total | 22.06 kB

Application bundle generation complete. [4.536 seconds] - 2026-07-06T11:26:29.494Z

Watch mode enabled. Watching for file changes...

NOTE: Raw file sizes do not reflect development server per-request transformations.

→ Local: <http://localhost:4200/>

→ press h + enter to show help

PS C:\Users\HP\Desktop\Deepskilling\Angular_HandsOn\student-course-portal> ng generate component pages/student-profile

Student Course Portal

[Home](#)[Courses](#)[Enroll](#)[Profile](#)[Notification](#)

Student Course Portal

Total Courses Available
3

Course Summary

Total Courses : 3

Add Course

Student Course Portal

[Home](#)[Courses](#)[Enroll](#)[Profile](#)[Notification](#)

Available Courses

SE

45

5

Passed

Add Course

Search courses

Search

Angular Fundamentals

Course Code: ANG101
Credits: 4
Status: pending

Unenroll

Edit

Delete

RxJS and State Management

Course Code: RXJ201
Credits: 3
Status: passed

Unenroll

Edit

Delete

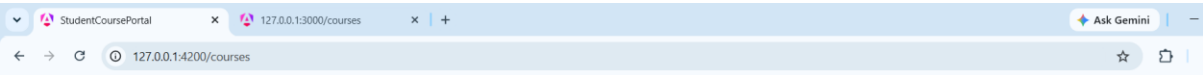
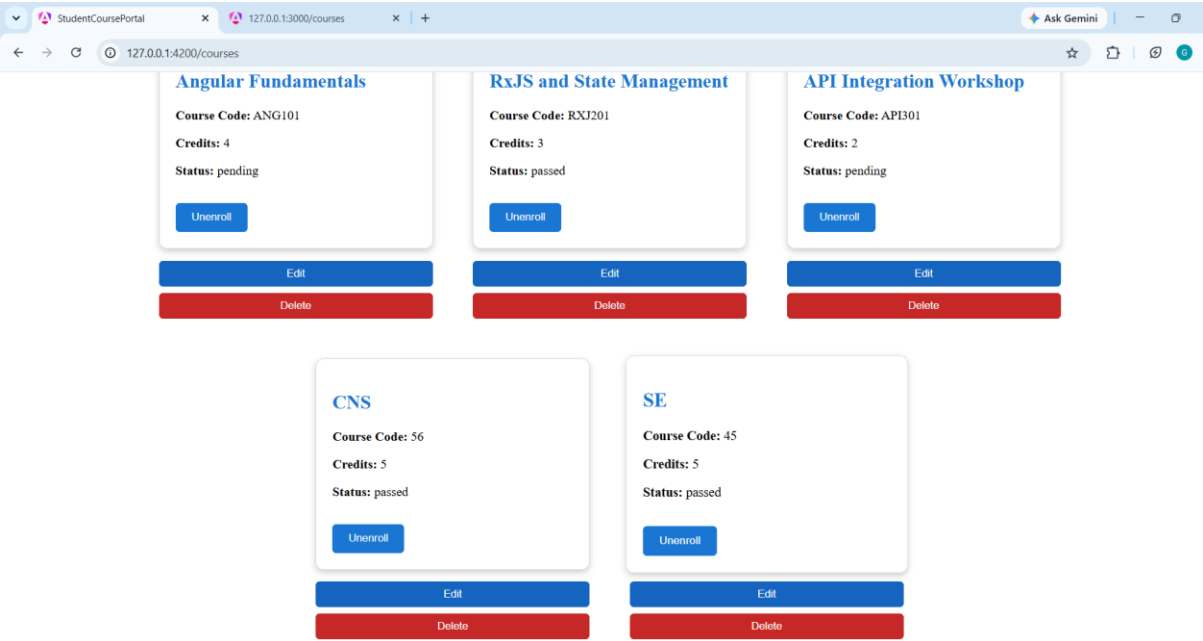
API Integration Workshop

Course Code: API301
Credits: 2
Status: pending

Unenroll

Edit

Delete



Student Course Portal

[Home](#) [Courses](#) [Enroll](#) [Profile](#) [Notification](#)

Available Courses

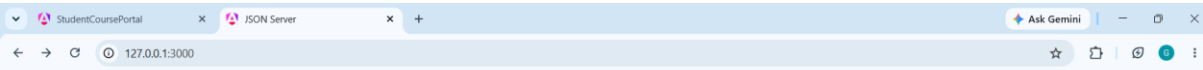
Course name Code 1 Pending Add Course

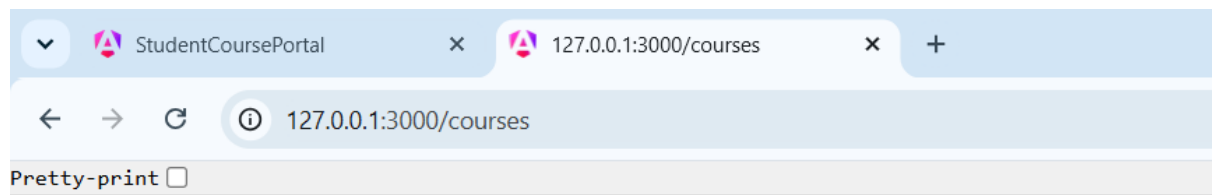
CNS Search

CNS
Course Code: 56
Credits: 5
Status: passed
Unenroll

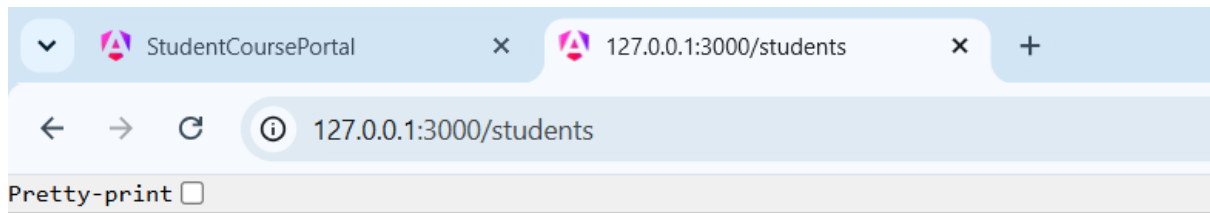
Edit

Delete

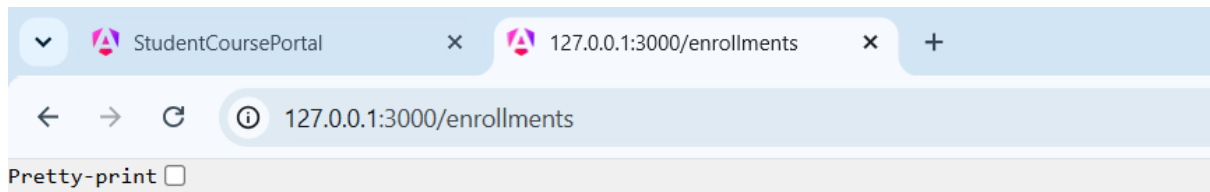




```
[
  {
    "id": "1",
    "name": "Angular Fundamentals",
    "code": "ANG101",
    "credits": 4,
    "gradeStatus": "pending"
  },
  {
    "id": "2",
    "name": "RxJS and State Management",
    "code": "RXJ201",
    "credits": 3,
    "gradeStatus": "passed"
  },
  {
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    "name": "API Integration Workshop",
    "code": "API301",
    "credits": 2,
    "gradeStatus": "pending"
  },
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    "credits": 5,
    "gradeStatus": "passed",
    "id": "Xt1a9hfflN8"
  },
  {
    "name": "SE",
    "code": "45",
    "credits": 5,
    "gradeStatus": "passed",
    "id": "PaAI3Gkd4Ww"
  }
]
```



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    "id": "1",
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    "email": "aarav.sharma@example.com"
  },
  {
    "id": "2",
    "name": "Meera Iyer",
    "email": "meera.iyer@example.com"
  },
  {
    "id": "3",
    "name": "Rohan Gupta",
    "email": "rohan.gupta@example.com"
  }
]
```



```
[
  {
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    "courseId": 1,
    "studentId": 1
  },
  {
    "id": "2",
    "courseId": 1,
    "studentId": 2
  },
  {
    "id": "3",
    "courseId": 2,
    "studentId": 1
  },
  {
    "id": "4",
    "courseId": 3,
    "studentId": 3
  }
]
```